SEMESTER IV (5 Papers)

Total 100 x 5 = 500 Marks

I. SKILL ENHANCEMENT COURSE SEC 2

(Credits: Theory-02) Marks : 75 (ESE: 3Hrs) + 25 (Pr 3Hrs) =100 Pass Marks ESE = 40

PHP PROGRAMMING THEORY: 15 LECTURES

1. INTRODUCTION TO PHP (3L)

PHP introduction, inventions and versions, important tools and software requirements (like Web Server, Database, Editors etc.)

PHP with other technologies, scope of PHP

Basic Syntax, PHP variables and constants

Types of data in PHP , Expressions, scopes of a variable (local, global)

PHP Operators : Arithmetic, Assignment, Relational , Logical operators, Bitwise , ternary and MOD operator.

PHP operator Precedence and associativity

2. HANDLING HTML FROM WITH PHP (2L)

Capturing Form Data

GET and POST form methods

Dealing with multi value fields

Redirecting a form after submission

3. PHP CONDITIONAL EVENTS AND LOOPS (2L)

PHP IF Else conditional statements (Nested IF and Else)

Switch case, while ,For and Do While Loop

Goto , Break ,Continue and exit

4. PHP FUNCTIONS: (3L)

Function, Need of Function , declaration and calling of a function

PHP Function with arguments, Default Arguments in Function

Function argument with call by value, call by reference

Scope of Function Global and Local

5. STRING MANIPULATION AND REGULAR EXPRESSION: (3L)

Creating and accessing String, Searching & Replacing String

Formatting, joining and splitting String, String Related Library functions

Use and advantage of regular expression over inbuilt function

Use of preg_match(), preg_replace(), preg_split() functions in regular expression Reference

6. ARRAY: (2L)

Anatomy of an Array, Creating index based and Associative array ,Accessing array

Looping with Index based array, with associative array using each() and foreach()

Some useful Library function

SKILL ENHANCEMENT LAB- SEC 2 LAB MARKS : PR (ESE: 3HRS)=25 PASS MARKS: PR (ESE) = 10

SOFTWARE LAB BASED ON PHP PROGRAMMING:

1. Create a PHP page using functions for comparing three integers and print the largest number.

2. Write a function to calculate the factorial of a number (non-negative integer). The function accept the number as an argument.

3. WAP to check whether the given number is prime or not.

4. Create a PHP page which accepts string from user. After submission that page displays the reverse of provided string.

5. Write a PHP function that checks if a string is all lower case.

6. Write a PHP script that checks whether a passed string is palindrome or not? (A palindrome is word, phrase, or sequence that reads the same backward as forward, e.g., madam or nurses run)

7. WAP to sort an array.

8. Write a PHP script that removes the whitespaces from a string.

a. Sample string : 'The quick " " brown

b. fox' Expected Output :

c. Thequick""brownfox

9. Write a PHP script that finds out the sum of first n odd numbers.

10. Create a login page having user name and password. On clicking submit, a welcome message should be displayed if the user is already registered (i.e.name is present in the database) otherwise error message should be displayed.

11. Write a simple PHP program to check that emails are valid

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III. CORE COURSE -C 8

DESIGN AND ANALYSIS OF ALGORITHMS

1. INTRODUCTION (5 LECTURES)

Basic Design and Analysis techniques of Algorithms, Correctness of Algorithm.

2. ALGORITHM DESIGN TECHNIQUES (8 LECTURES)

Iterative techniques, Divide and Conquer, Dynamic Programming, Greedy Algorithms.

3. SORTING AND SEARCHING TECHNIQUES (20 LECTURES)

Elementary sorting techniques–Bubble Sort, Insertion Sort, Merge Sort, Advanced Sorting techniques - Heap Sort, Quick Sort, Sorting in Linear Time - Bucket Sort, Radix Sort and Count Sort, Searching Techniques, Medians & Order Statistics, complexity analysis;

4. LOWER BOUNDING TECHNIQUES (5 LECTURES)

Decision Trees

5. BALANCED TREES (7 LECTURES)

Red-Black Trees

6. ADVANCED ANALYSIS TECHNIQUE (5 LECTURES)

Amortized analysis

7. GRAPHS (5 LECTURES)

Graph Algorithms–Breadth First Search, Depth First Search and its Applications, Minimum Spanning Trees.

COMPUTER PRACTICAL- C 8 LAB

DESIGN AND ANALYSIS OF ALGORITHMS

1. i. Implement Insertion Sort (The program should report the number of comparisons)

ii. Implement Merge Sort (The program should report the number of comparisons)

2. Implement Heap Sort(The program should report the number of comparisons)

3. Implement Randomized Quick sort (The program should report the number of comparisons)

4. Implement Radix Sort

5. Create a Red-Black Tree and perform following operations on it:

i. Insert a node

ii. Delete a node

iii. Search for a number & also report the color of the node containing this number.

6. Write a program to determine the LCS of two given sequences

7. Implement Breadth-First Search in a graph

8. Implement Depth-First Search in a graph

9. Write a program to determine the minimum spanning tree of a graph

IV. CORE COURSE -C 9: (CREDITS: THEORY-04, PRACTICALS-02)

1. INTRODUCTION (8 LECTURES)

The Evolving Role of Software, Software Characteristics, Changing Nature of Software, Software

Engineering as a Layered Technology, Software Process Framework, Framework and Umbrella

Activities, Process Models, Capability Maturity Model Integration (CMMI).

2. REQUIREMENT ANALYSIS (10 LECTURES)

Software Requirement Analysis, Initiating Requirement Engineering Process, Requirement

Analysis and Modeling Techniques, Flow Oriented Modeling, Need for SRS, Characteristics and

Components of SRS.

3. SOFTWARE PROJECT MANAGEMENT (8LECTURES)

Estimation in Project Planning Process, Project Scheduling.

4. RISK MANAGEMENT (8 LECTURES)

Software Risks, Risk Identification, Risk Projection and Risk Refinement, RMMM Plan.

5. QUALITY MANAGEMENT (8 LECTURES)

Quality Concepts, Software Quality Assurance, Software Reviews, Metrics for Process and Projects.

6. DESIGN ENGINEERING (10 LECTURES)

Design Concepts, Architectural Design Elements, Software Architecture, Data Design at the Architectural Level and Component Level, Mapping of Data Flow into Software Architecture, Modeling Component Level Design.

7. TESTING STRATEGIES & TACTICS (8 LECTURES)

Software Testing Fundamentals, Strategic Approach to Software Testing, Test Strategies for Conventional Software, Validation Testing, System testing, Black-Box Testing, White-Box Testing and their type, Basis Path Testing.

COMPUTER PRACTICAL-C 9 LAB 60 LECTURES

Understanding SRS (Elements of SRS), Creation of SRS according to IEEE standard.

Case Study of

1. Online Retail Shopping Management System.
2. Online Hotel Reservation System.
3. Examination & Result Computation System.
4. Automatic Internal Assessment System.
5. Parking allocation system.
6. Whole Sale Management System

V. CORE COURSE -C 10

DATABASE MANAGEMENT SYSTEMS THEORY: 60 LECTURES

1.INTRODUCTION (6 LECTURES)

Characteristics of database approach, data models, database system architecture and data independence.

2. 2.ENTITY RELATIONSHIP(ER) MODELING (8 LECTURES)

Entity types, relationships, constraints.

3. 3.RELATION DATA MODEL (20 LECTURES)

Relational model concepts, relational constraints, relational algebra, SQL queries

4. 4.DATABASE DESIGN (15 LECTURES)

Mapping ER/EER model to relational database, functional dependencies, Lossless decomposition, Normal forms (upto BCNF).

5. TRANSACTION PROCESSING (3 LECTURES)

ACID properties, concurrency control

6. 6. FILE STRUCTURE AND INDEXING (8 LECTURES)

Operations on files, File of Unordered and ordered records, overview of File organizations, Indexing structures for files(Primary index, secondary index, clustering index), Multilevel indexing using B and B+ trees.

COMPUTER PRACTICAL-C 10 LAB 60 LECTURES

DATABASE MANAGEMENT SYSTEMS

Create and use the following database schema to answer the given queries.

EMPLOYEE Schema

Field Type NULL KEY DEFAULT

Eno Char(3) NO PRI NIL

Ename Varchar(50) NO NIL

Job_type Varchar(50) NO NIL

Manager Char(3) Yes FK NIL

Hire_date Date NO NIL

Dno Integer YES FK NIL

Commission Decimal(10,2) YES NIL

Salary Decimal(7,2) NO NIL

DEPARTMENT Schema

Field Type NULL KEY DEFAULT

Dno Integer No PRI NULL

Dname Varchar(50) Yes NULL

Location Varchar(50) Yes New Delhi

Query List

SECTION-I

1. Query to display Employee Name, Job, Hire Date, Employee Number; for each employee with the

Employee Number appearing first.

2. Query to display unique Jobs from the Employee Table.

3. Query to display the Employee Name concatenated by a Job separated by a comma.

4. Query to display all the data from the Employee Table. Separate each Column by a comma and

name the said column as THE_OUTPUT.

5. Query to display the Employee Name and Salary of all the employees earning more than $2850.

6. Query to display Employee Name and Department Number for the Employee No= 7900.

7. Query to display Employee Name and Salary for all employees whose salary is not in the range of

$1500 and $2850.

8. Query to display Employee Name and Department No. of all the employees in Dept 10 and Dept

30 in the alphabetical order by name.

9. Query to display Name and Hire Date of every Employee who was hired in 1981.

10. Query to display Name and Job of all employees who don‘t have a current Manager.

SECTION-II

1. Query to display the Name, Salary and Commission for all the employees who earn commission.

2. Sort the data in descending order of Salary and Commission.

3. Query to display Name of all the employees where the third letter of their name is _A‘.

4. Query to display Name of all employees either have two _R‘s or have two _A‘s in their name and are either in Dept No = 30 or their Manger‘s Employee No = 7788.

5. Query to display Name, Salary and Commission for all employees whose Commission Amount is 14 greater than their Salary increased by 5%.

6. Query to display the Current Date.

7. Query to display Name, Hire Date and Salary Review Date which is the 1st Monday after six months of employment.

8. Query to display Name and calculate the number of months between today and the date each employee was hired.

9. Query to display the following for each employee <E-Name> earns < Salary> monthly but wants < 3 * Current Salary >. Label the Column as Dream Salary.

10. Query to display Name with the 1st letter capitalized and all other letter lower case and length of their name of all the employees whose name starts with _J‘, ‘A‘ and _M‘.

GEOGRAPHY GENERIC ELECTIVE 4

IV. GENERIC ELECTIVE (GE 4)

ECONOMIC GEOGRAPHY Theory: 60 Lectures

1. Introduction: Concept and classification of economic activity

2. Factors Affecting location of Economic Activity with special reference to Agriculture (Von Thunen theory), Industry (Weber’s theory).

3. Primary Activities: Subsistence and Commercial agriculture, forestry, fishing and mining.

4. Secondary Activities: Manufacturing (Cotton Textile, Iron and Steel- with reference to world),

Concept of Manufacturing Regions, Special Economic Zones

5. Tertiary Activities: Transport, Trade and Services.

GEOGRAPHY PRACTICAL- GE4 LAB

INSTRUMENTAL SURVEY

1. Plane Table: Radiation, Intersection Method.

2. Prismatic Compass: Open & Closed Traverse.

SEMESTER IV GENERIC ELECTIVE

MONEY BANKING AND PUBLIC FINANCE

1. MONEY

Concept, functions, measurement; theories of money supply and determination.

2. FINANCIAL INSTITUTIONS, MARKETS, INSTRUMENTS AND FINANCIAL INNOVATIONS

(a) Role of financial markets and institutions; financial crisis.

(b) Money and capital markets: organization, structure and reforms in India;

3. INTEREST RATES

Determination; source of interest rate differentials

4. INDIAN BANKING SYSTEM

Changing role and structure; banking sector reforms.

5. CENTRAL BANKING AND MONETARY POLICY

Functions, balance sheet, goals, targets, indicators and instruments of monetary control; monetary management in an open economy; current monetary policy of India.

6. THEORY OF TAXATION

Ability to pay, Benefit principle, Canons of taxation, shifting, Incidence and of Taxes.

7. MONETARY AND FISCAL POLICIES

Direct-Tax, Indirect-Tax,(GST).Goods and Services Tax, Analysis of Budget.